

GIBBS DIFFUSION FOR BLIND DENOISING

GENERATIVE MODELING FOR BLIND DENOISING AND SIMULTANEOUS NOISE PARAMETER ESTIMATION

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Motivation & Scientific Justification

The extraction of a clean signal x from a noisy observation y is a ubiquitous challenge. In the "blind denoising" problem, both the signal and the noise characteristics are unknown, making it particularly ill-posed. This project reproduces the Gibbs Diffusion (GDiff) framework by Heurtel-Depeiges et al. (2024) [1], a novel Bayesian approach that addresses this challenge. GDiff aims to simultaneously infer the clean signal and the parameters ϕ of a corrupting noise process $\varepsilon \sim \mathcal{N}(0, \Sigma_\phi)$. We validate GDiff's efficacy on two distinct applications: denoising natural images corrupted by unknown colored noise, and inferring cosmological parameters from simulated Cosmic Microwave Background (CMB) data. Providing a full posterior distribution, rather than a single point estimate, is critical for robust uncertainty quantification in sensitive scientific analyses.

Bayesian Framework & Methodology

GDiff operates within a Bayesian framework to sample from the joint posterior distribution $p(x, \phi | y)$. According to Bayes' theorem, this posterior is proportional to the product of the likelihood and the priors:

$$p(x, \phi | y) \propto \underbrace{p(y | x, \phi)}_{\text{Likelihood}} \underbrace{p(x)}_{\text{Signal Prior}} \underbrace{p(\phi)}_{\text{Noise Prior}}$$

Here, the likelihood is derived from the Gaussian noise model, $p(y | x, \phi) = \mathcal{N}(y; x, \Sigma_\phi)$. The key innovation of GDiff is that the signal prior $p(x)$ is not analytical, but is implicitly defined by a powerful, pre-trained conditional diffusion model [2]. This deep generative model learns the complex, high-dimensional distribution of natural signals from a large dataset, providing a much stronger and more flexible prior than traditional analytical forms (e.g., sparsity).

Since direct sampling from this joint posterior is intractable, GDiff employs a Gibbs sampler [5] that iteratively draws from the two full conditional distributions:

1. **Signal Denoising (Diffusion Step):** In this step, a new signal estimate $x^{(k)}$ is sampled from its conditional posterior:

$$x^{(k)} \sim p(x | y, \phi^{(k-1)})$$

This sampling is approximated by running the reverse process of a conditional diffusion model, which is trained to denoise an observation y given a specific noise parameterization ϕ . This effectively leverages the learned signal prior $p(x)$ to find a plausible clean signal consistent with the observation and the current noise estimate.

2. **Noise Parameter Inference (HMC Step):** Next, new noise parameters $\phi^{(k)}$ are sampled from their conditional posterior:

$$\phi^{(k)} \sim p(\phi | y, x^{(k)})$$

Given the current signal estimate $x^{(k)}$, the noise realization is computed as $\varepsilon^{(k)} = y - x^{(k)}$. This simplifies the problem to sampling from $p(\phi | \varepsilon^{(k)})$. This sampling is performed using Hamiltonian Monte Carlo (HMC) [4], an efficient MCMC method. HMC explores the target distribution by simulating Hamiltonian dynamics on a potential energy surface $U(\phi) = -\log p(\phi | \varepsilon^{(k)})$. The full negative log-posterior (potential energy) is given by:

$$U(\phi) = \frac{1}{2}(\varepsilon^{(k)})^T \Sigma_\phi^{-1} \varepsilon^{(k)} + \frac{1}{2} \log |\Sigma_\phi| - \log p(\phi)$$

where the gradient $\nabla_\phi U(\phi)$ is required for the HMC updates.

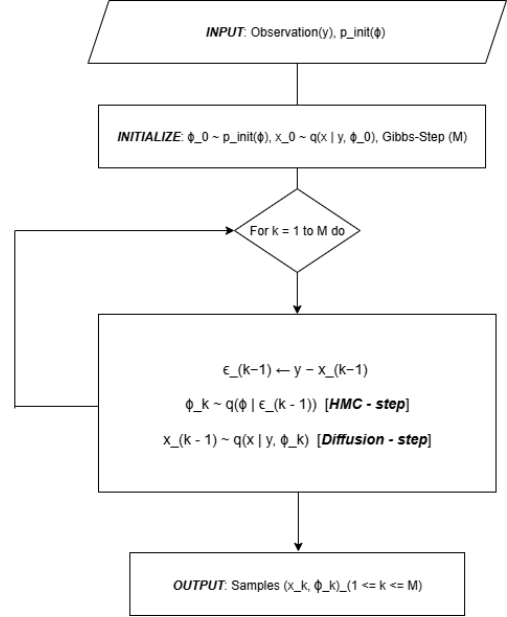


Figure 1: The GDiff Gibbs sampling loop, alternating between signal sampling via diffusion and noise parameter sampling via HMC.

Key Findings

This reproduction study successfully validated the core claims of GDiff, while also revealing practical challenges.

1. **Blind Denoising of Natural Images:** GDiff effectively denoised images corrupted by colored noise, where the noise power spectrum is $S(k) \propto |k|^{\bar{\phi}}$. The posterior mean estimate $\mathbb{E}[x|y]$ consistently outperformed the BM3D baseline [6] in quantitative metrics (Table 1). The inference on the noise amplitude σ was accurate, but the posterior for the spectral index $\bar{\phi}$ showed a slight bias (e.g., mean of 0.95 vs. true 1.0) due to computationally constrained chain lengths (Figure 2).

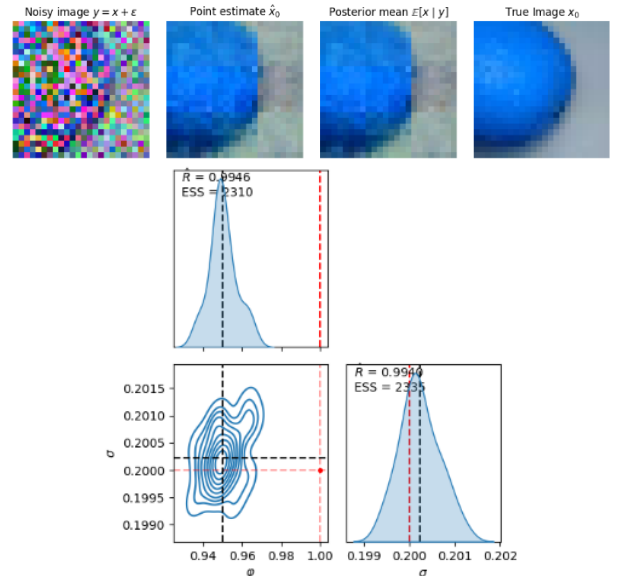


Figure 2: Example of blind denoising ($\sigma = 0.2, \bar{\phi} = 1.0$). Upper: visual comparison. Lower: inferred posteriors for $(\sigma, \bar{\phi})$ with true values marked by red lines.

Table 1: Example PSNR (dB) / SSIM for Tiny-ImageNet, $\sigma = 0.20$.

Noise Type (ϕ)	BM3D	GDiff $\mathbb{E}[x y]$
Pink (-1)	22.0 / 0.45	23.8 / 0.59
White (0)	22.7 / 0.52	23.9 / 0.60
Blue (1)	22.6 / 0.48	23.4 / 0.59

2. **Cosmological Parameter Inference:** GDiff was applied to separate Galactic dust (signal) from CMB (noise), where $\phi = (H_0, \omega_b, \sigma_{\text{CMB}})$. While visual component separation was successful, severe computational constraints (e.g., only 5 Gibbs steps) and kernel crashes during likelihood evaluation via CAMB [8] prevented the HMC sampler from converging. This resulted in poorly formed posteriors with abnormal convergence diagnostics ($\hat{R} \gg 1$), even as the mean of some parameters like H_0 approached the true value (Figure 3).

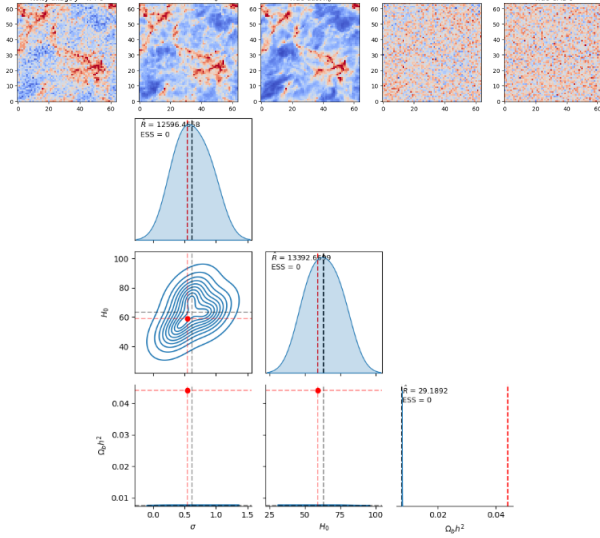


Figure 3: Upper: Successful visual separation of dust and CMB. Lower: Poorly converged posteriors for cosmological parameters due to computational limits. ($\sigma_{\text{CMB}} = 0.54$, $H_0 = 59.14$, $\omega_b = 0.044$)

Challenges & Conclusion

The primary obstacle was the immense computational cost of the iterative inference pipeline. Training the diffusion models and running numerous Gibbs/HMC chains was resource-intensive, and instability on CSD3 cluster nodes severely limited the achievable chain lengths.

This directly impacted the quality of posterior inference, especially for the complex cosmological application.

This reproduction study confirms Gibbs Diffusion as a powerful and theoretically grounded framework for Bayesian blind denoising. The implementation successfully replicated GDiff’s ability to co-infer signals and noise parameters, providing high-quality visual reconstructions and competitive quantitative results. However, its significant computational intensity poses a major practical barrier to achieving well-converged posteriors in complex problems under typical resource constraints. GDiff’s methodology is sound, but its practical application necessitates substantial computational resources or future work on algorithmic efficiency.

References

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